

Census Migration Flow Descriptive Analysis

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1. Top Destinations

Table: Top Destination Regions

Region	Average Inflow as Population Share
Norte Mato-grossense	28.0
Sul de Roraima	21.5
Leste Rondoniense	17.3
Leste Goiano	17.3
Sudeste Paraense	17.1
Nordeste Mato-grossense	14.0
Madeira-Guaporé	13.3
Sudoeste Paraense	13.0
Distrito Federal	12.9
Norte de Roraima	12.9

- In absolute terms population totals dominates any effect
- Population shares – the agri frontier is the main receiver

2. Top Origins

Table: Top Origin Regions

Region	Average Outflow as Population Share
Centro Ocidental Paranaense	19.6
Noroeste Paranaense	16.2
Noroeste Goiano	14.1
Norte Pioneiro Paranaense	13.5
Norte Goiano	13.2
Oeste Paranaense	13.2
Sudoeste Paranaense	12.6
Vale do Mucuri	12.3
Sudoeste Mato-grossense	11.6
Noroeste Espírito-santense	11.2

- Heavy Paraná out-migration was unexpected!

3. Effective Origins and Destinations

Table: HHI implied linkages

index	Effective Origins	Effective Destinations
count	137.0	137.0
mean	9.5	9.0
std	5.7	4.9
min	2.5	1.7
255075max	47.1	34.0

- The median region receives from/sends to around 8.5 out of 137 regions

4. O-D Persistence

Table: Persistence in Origins and Destinations

Decade	Origin Outflow Autocorrelation	Destination Inflow Autocorrelation
1990	89.2	96.2
2000	95.6	96.5
2010	97.7	98.2

- Destination persistence is very high
- origin persistence is increasing!

5. Migration Corridors

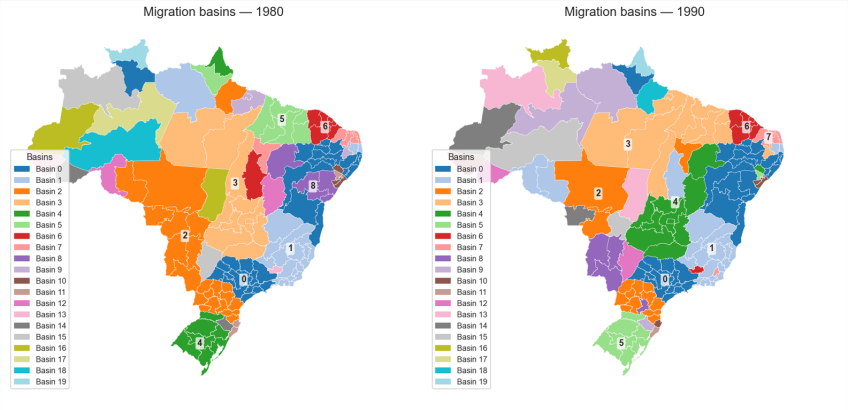
Table: Migration Corridor Analysis

Decade	Effective Corridors	Flow Autocorrelation	Average Survival Rate
1990	744.3	80.0	80.2
2000	706.7	79.4	88.2
2010	846.1	79.5	85.9

- Out of over 18 thousand corridors, only around 700 are effectively active per decade
- Corridor persistence is high

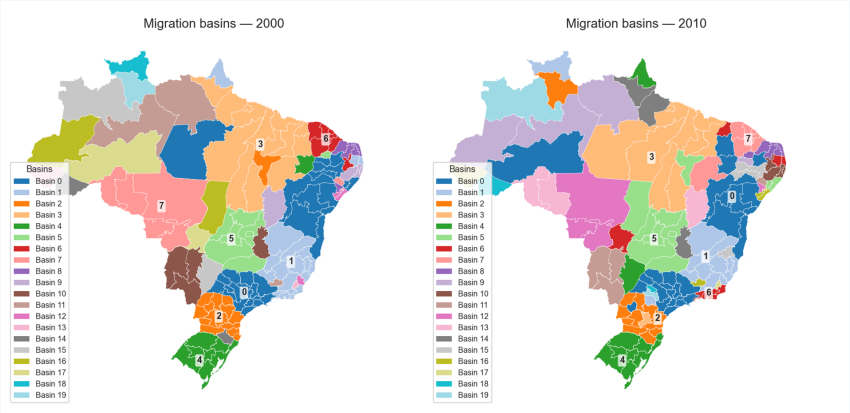
6.1 Apparent Migration Basins – 1980/90

Figure: Migratory Relationships per Decade



6.2 Apparent Migration Basins – 2000/10

Figure: Migratory Relationships per Decade



7. Reciprocity and Redistribution

Table: Distributive Effects of Migrations

Decade	Reciprocity	One-Wayness	Redistribution Index
198000	52.9	47.1	32.5
199000	59.8	40.2	23.8
200000	65.7	34.3	20.5
201000	69.2	30.8	18.7

- $\text{reciprocity} = 2 \min(\text{flow}_{ij}, \text{flow}_{ji}) / (\text{flow}_{ij} + \text{flow}_{ji})$
- $\text{one wayness} = |\text{flow}_{ij} - \text{flow}_{ji}| / (\text{flow}_{ij} + \text{flow}_{ji})$
- Flows become less redistributive with time

8. Migration vs. Climate and GDP per Capita

Attribute	Exit Rate Correlation
Climate Exposure	-0.023
GDP per Capita	-0.35

GDP per Capita			
Origin Tercile	Destination Tercile		
	Low	Medium	High
Low	0.07	0.08	0.13
Medium	0.04	0.06	0.15
High	0.06	0.10	0.32

Climate Exposure			
Origin Tercile	Destination Tercile		
	Low	Medium	High
Low	0.17	0.12	0.05
Medium	0.13	0.135	0.08
High	0.08	0.097	0.14

- Exit rates correlate strongly with GDP per capita, weakly with climate
- Migration follows low-to-high GDP per capita logic
- Climate is muddier: highly exposed regions migrate to other highly exposed regions

Proposed Continuations

- Ponticelli et al study labor relocation in response to dry spells and frame migration as a climate adaptation policy
- Just getting the elasticities right and looking at welfare sounds derivative
- More original: Study the impact of migration on those who stay / "trapped populations"
- Issue: how to differentiate welfare of stayers and migrants from a same origin?
- Less translatable skills? Financial restrictions?